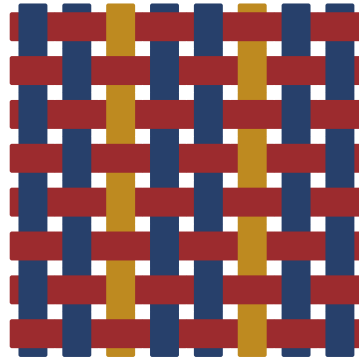




Gradient Descent

Rolling downhill, and the price of a bad bowl · fill-in notes

woven on Loom

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dyed in indigo, madder & weld; ruled in iron-gall. woven on Loom.

THE ONE IDEA, AND HOW TO READ THESE NOTES

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To minimize f , stand on its graph and walk downhill: $x_{k+1} = x_k - \eta \nabla f(x_k)$. How fast you reach the bottom is set entirely by the **shape of the bowl**. The single idea of these notes: the convergence rate is a **tax on conditioning**. A round bowl is paid off fast; a long narrow valley slowly; and every acceleration trick, from momentum to preconditioning, is a way to pay less of that tax.

The master table: the rate, and what sets it (fill it in as you go)

assumptions on f	rate / steps to error ε
convex, L -smooth	$f(x_k) - f^* = O(1/k)$, so $O(\rule{1cm}{0.4pt})$ steps (§2)
μ -strongly convex, L -smooth	linear: $(1 - 1/\kappa)^k$, so $O(\kappa \log \frac{1}{\varepsilon})$ (§3)
condition number	$\kappa = \rule{1cm}{0.4pt}$ (the elongation of the bowl)
accelerated (Nesterov)	$O(\rule{1cm}{0.4pt} \log \frac{1}{\varepsilon})$: the optimal first-order rate (§4)
stochastic (SGD)	cheap steps, but a $\rule{1cm}{0.4pt}$ floor

How to read (passive *and* active)

- **Read** the strands and the stated theorems.
- **Fill** the blanks $\rule{1cm}{0.4pt}$, the [fill in: missing step] proof gaps, and the *Your turn* boxes. Any convex-optimization text is the answer key.
- Bump each $\backslash\mathrm{warmth}\{0\}$ to 0–5 once you can rebuild it cold.

CONTENTS

The one idea · how to read	1
1 The method	3
2 Convex and smooth: guaranteed descent	3
3 Conditioning: the tax, named	4
4 Paying the tax down	5

1 THE METHOD

□□□□ The gradient points uphill. Negate it, take a step, repeat.

At a point x , the gradient $\nabla f(x)$ is the direction of *steepest ascent*, and its negative the steepest descent. So the greedy thing to do is step a little way against the gradient and reassess. The only knob is the **step size** η : too small and you crawl, too large and you overshoot the valley and may climb the far wall.

The update

Definition 1.1 (*gradient descent*). Starting from x_0 , iterate

$$x_{k+1} = x_k - \eta \text{ _____ },$$

with step size $\eta > 0$. The fixed points are exactly the **stationary points** $\nabla f = 0$; for convex f those are the global minimizers.

Your turn. Test it on a one-dimensional bowl $f(x) = \frac{1}{2} a x^2$ with $a > 0$.

- The update becomes $x_{k+1} = (1 - \eta a) x_k$, so $x_k = \text{_____} \cdot x_0$.
- It converges to 0 exactly when $|1 - \eta a| < 1$, i.e. for $0 < \eta < \text{_____}$. The sweet spot $\eta = 1/a$ lands in *one* step.

That toy already shows the whole story. The good step size is tied to the **curvature** a : match it and you fly, exceed $2/a$ and you diverge. In many dimensions there are *many* curvatures at once (the eigenvalues of the Hessian), and a single η must serve all of them. The spread of those curvatures is the tax of the next two sections.

2 CONVEX AND SMOOTH: GUARANTEED DESCENT

□□□□ Convexity rules out traps; smoothness caps the curvature; together they force $O(1/k)$.

Two assumptions make the toy rigorous in any dimension. **Convexity** says the bowl has no false bottoms, every stationary point is a global minimum. **L-smoothness** says the gradient never changes faster than rate L , so a step of size $1/L$ is always safe. Together they guarantee the value falls every step, and reaches the minimum at rate $O(1/k)$.

? Why $-\nabla f$ and not some other descent direction?
Among all unit directions d , the first-order change $\langle \nabla f(x), d \rangle$ is most negative when $d \propto -\nabla f(x)$. Steepest descent is exact here because the curvature is constant.

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If  $f$  is not differentiable (a kink, an  $\ell_1$  penalty), replace  $\nabla f$  by a **subgradient**; the method still works, just more slowly, and the proximal variants of §4 handle the kink exactly.

## The two assumptions

- **Definition 2.1** (*convex, L-smooth*).  $f$  is **convex** if it lies above each tangent:  $f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle$ . It is **L-smooth** if  $\nabla f$  is L-Lipschitz, equivalently it lies below each tangent parabola:

$$f(y) \leq f(x) + \langle \nabla f(x), y - x \rangle + \frac{L}{2} \|y - x\|^2.$$

## Descent, then the rate

- **Lemma 2.2** (*descent lemma*). With step  $\eta = 1/L$ , a gradient step strictly decreases  $f$ :

$$f(x_{k+1}) \leq f(x_k) - \frac{1}{2L} \|\nabla f(x_k)\|^2.$$

*Proof.* Put  $y = x_{k+1} = x_k - \frac{1}{L} \nabla f(x_k)$  into the smoothness inequality. [ fill in: expand the two terms; the cross term  $-\frac{1}{L} \|\nabla f\|^2$  and  $+\frac{1}{2L} \|\nabla f\|^2$  combine to  $-\frac{1}{2L} \|\nabla f\|^2$  ] ■

- **Theorem 2.3** ( *$O(1/k)$  for convex, smooth*). If  $f$  is convex and L-smooth, then gradient descent with  $\eta = 1/L$  satisfies

$$f(x_k) - f^* \leq \frac{L \|x_0 - x^*\|^2}{2k}.$$

*Proof.* Skeleton: convexity bounds the gap  $f(x_k) - f^* \leq \langle \nabla f(x_k), x_k - x^* \rangle$ ; combine with the descent lemma and telescope the distances  $\|x_k - x^*\|^2$ . [ fill in: carry out the telescoping sum and divide by  $k$  ] ■

**Your turn.** Read the rate as a budget. To reach  $f(x_k) - f^* \leq \varepsilon$ , Theorem 2.3 needs

$$k = O(\text{_____}) \text{ steps.}$$

Halving  $\varepsilon$  \_\_\_\_\_ the number of steps.

A note on  $L$ : it is the largest eigenvalue of the Hessian (for a quadratic,  $L = \lambda_{\max}$ ), so the safe step  $1/L$  is set by the steepest curvature. If other directions are much flatter, that one step size wastes their progress, exactly the complaint §3 makes precise.

?  $k = O(1/\varepsilon)$ ; halving  $\varepsilon$  doubles the steps. This "sublinear" rate is honest but slow, which is exactly what strong convexity will fix.

## ■ 3 CONDITIONING: THE TAX, NAMED

□□□□ Curve the bowl up everywhere and descent goes geometric. The rate is set by  $\kappa = L/\mu$ .

Add one more assumption, **strong convexity**: the bowl curves up at rate at least  $\mu$  everywhere, so it has a unique minimum and walls that push back. Now gradient descent converges **linearly** (the error shrinks by a constant factor each step), and the factor is governed by a single number, the **condition number**  $\kappa = L/\mu$ . This is the tax.

## Strong convexity and the linear rate

**Definition 3.1** ( $\mu$ -strong convexity).  $f$  is  **$\mu$ -strongly convex** if it lies above each tangent parabola:  $f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle + \frac{\mu}{2} \|y - x\|^2$ . Sandwiched with  $L$ -smoothness, the curvature is pinned into  $[\mu, L]$ .

**Theorem 3.2** (linear convergence). If  $f$  is  $\mu$ -strongly convex and  $L$ -smooth, gradient descent with  $\eta = 1/L$  obeys

$$\|x_k - x^*\|^2 \leq \left(1 - \frac{\mu}{L}\right)^k \|x_0 - x^*\|^2 = \left(1 - \frac{1}{\kappa}\right)^k \|x_0 - x^*\|^2.$$

*Proof.* Skeleton: show one step is a contraction,  $\|x_{k+1} - x^*\| \leq (1 - \mu/L)^{1/2} \|x_k - x^*\|$ , using strong convexity and smoothness to bound  $\langle \nabla f(x_k), x_k - x^* \rangle$  from below. [ fill in: expand  $\|x_k - \eta \nabla f(x_k) - x^*\|^2$  and apply the two curvature bounds ]

Picture the level sets. With curvatures in  $[\mu, L]$  the bowl is an ellipse whose axis ratio is  $\kappa = L/\mu$ . When  $\kappa = 1$  it is a perfect circle and descent points straight at the minimum, done in one step. When  $\kappa$  is large it is a long thin valley: the gradient points mostly across the valley, not along it, so the iterates **zigzag** and crawl toward the bottom. Ill-conditioning, not dimension, is what makes gradient descent slow.

**Your turn.** Turn the rate into a step count. Since  $(1 - 1/\kappa)^k \leq e^{-k/\kappa}$ ,

$$k = O\left(\kappa \log \frac{1}{\epsilon}\right) \text{ steps for error } \epsilon.$$

- A round bowl  $\kappa = 1$  needs about \_\_\_\_\_ steps; a valley with  $\kappa = 10^4$  needs about \_\_\_\_\_.
- For a quadratic,  $\kappa$  is the ratio  $\lambda_{\max}/\lambda_{\min}$  of the Hessian, i.e. its \_\_\_\_\_.

Compare §2: convex-only gives  $O(1/\epsilon)$ , strongly convex gives  $O(\kappa \log \frac{1}{\epsilon})$ . The log is a huge win (linear vs. sublinear); the  $\kappa$  in front is the price still to be haggled down in §4.

## 4 PAYING THE TAX DOWN

□□□□ Momentum buys  $\sqrt{\kappa}$ ; stochasticity buys cheap steps; Newton erases  $\kappa$  at a price.

?  $O(\log \frac{1}{\epsilon})$  vs.  $O(10^4 \log \frac{1}{\epsilon})$ ; and  $\kappa$  is the Hessian's *condition number* (the SVD's  $\sigma_{\max}/\sigma_{\min}$ ). The valley costs you a factor of  $\kappa$ .

The condition number  $\kappa$  is the bill. The art of optimization is haggling it down. Three moves dominate: **momentum** (remember where you were going, and reduce  $\kappa$  to  $\sqrt{\kappa}$ ), **stochasticity** (use a cheap noisy gradient on huge data), and **preconditioning / Newton** (reshape the bowl to a circle, erasing  $\kappa$  entirely).

## The landscape of rates

| method / setting      | steps to $\varepsilon$                        | what it buys                                                      |
|-----------------------|-----------------------------------------------|-------------------------------------------------------------------|
| GD, convex            | $O(1/\varepsilon)$                            | baseline, sublinear                                               |
| GD, strongly convex   | $O(\kappa \log \frac{1}{\varepsilon})$        | linear, but taxed by $\kappa$                                     |
| Nesterov accelerated  | $O(\sqrt{\kappa} \log \frac{1}{\varepsilon})$ | <b>optimal</b> first-order rate                                   |
| SGD (strongly convex) | $O(1/\varepsilon)$                            | each step $O(1)$ not $O(n)$ ; noise floor needs decreasing $\eta$ |
| GD, nonconvex smooth  | $O(1/\varepsilon^2)$                          | only a <i>stationary</i> point $\ \nabla f\  \leq \varepsilon$    |

**Theorem 4.1** (*acceleration*). For  $\mu$ -strongly convex,  $L$ -smooth  $f$ , Nesterov's momentum method converges as  $(1 - 1/\sqrt{\kappa})^k$ , needing only  $O(\sqrt{\kappa} \log \frac{1}{\varepsilon})$  steps. No first-order method (one gradient per step) can beat  $\sqrt{\kappa}$  on the worst such  $f$ .

**Your turn.** Count the payoff on a stiff problem,  $\kappa = 10^4$  (ignore the log factor).

plain GD:  $\approx$  \_\_\_\_\_ steps,      accelerated:  $\approx$  \_\_\_\_\_ steps,      speedup  $\approx$  \_\_\_\_\_.

## The one idea, recovered

Read the master table again. Every rate is the same story told at a different conditioning. A round bowl ( $\kappa = 1$ ) is trivial; the tax grows with  $\kappa$ ; momentum softens it to  $\sqrt{\kappa}$ ; a good **preconditioner** (in the limit, Newton's method, which rescales by the inverse Hessian) reshapes the bowl back to a circle and removes  $\kappa$  altogether, at the cost of solving a linear system each step. **The rate is a tax on conditioning, and optimization is the art of paying less.**

? Momentum's trick: extrapolate a little past the current point before taking the gradient, so consistent progress builds speed and the cross-valley zigzag cancels. It is the momentum is matching upper bound to the genuine  $\sqrt{\kappa}$  lower bound and not.

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The machine-learning reality: f is nonconvex and n is enormous, so practice is SGD with momentum and **adaptive** per-coordinate steps (Adam) that cheaply approximate preconditioning. The same three levers, conditioning, momentum, stochasticity, are still exactly what is being traded.